

Java App Deployment Flow: Maven -> Docker -> Docker Compose

1. Maven Build (mvn package)

This compiles your code and builds the .jar file:

```
mvn clean package
```

Result: Creates target/your-app.jar (e.g., target/employee-management-docker.jar)

2. Docker Build

You use a Dockerfile to containerize the application.

Example Dockerfile:

```
FROM openjdk:17-jdk-slim
WORKDIR /app
ADD target/employee-management-docker.jar app.jar
ENTRYPOINT ["java", "-jar", "app.jar"]
```

Then build the Docker image:

```
docker build -t springemployeemanagement:1.1 .
```

Result: Docker image tagged as springemployeemanagement:1.1

3. Docker Compose (Optional)

For running multiple services or simplifying local setup.

Example docker-compose.yml:

```
version: '3.8'

services:

  employee-service:

    image: springemployeemanagement:1.1

    build:

      context: .

      dockerfile: Dockerfile

    ports:

      - "8080:8080"

    imagePullPolicy: never
```

Run using Docker Compose:

```
docker-compose up --build
```

Result: Your container is running and accessible at localhost:8080

Summary Flow

Maven build ---> Creates .jar

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Docker build ---> Uses Dockerfile to create image

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Docker Compose ---> Optionally runs multiple containers (DB, API, etc.)